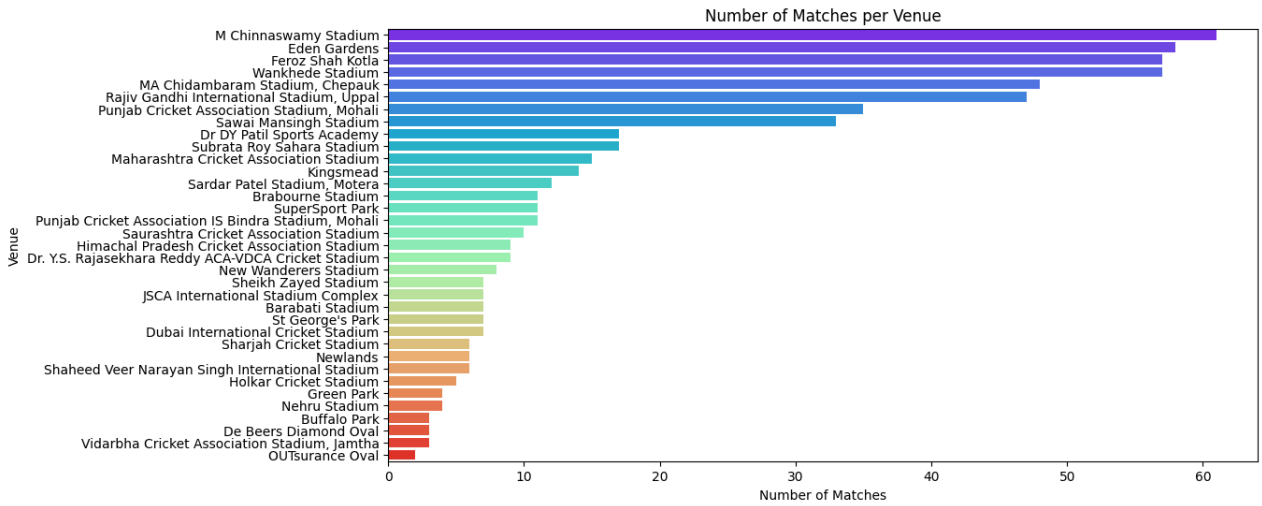
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn import preprocessing
import keras
import tensorflow as tf
```

```
ipl = pd.read_csv('ipl_data.csv')
ipl.head()
```

	mid	date	venue	bat_team	bowl_team	batsman	bowler	runs	wickets	overs	runs_last_5	wickets_last_5
0	1	2008-04-18	M Chinnaswamy Stadium	Kolkata Knight Riders	Royal Challengers Bangalore	SC Ganguly	P Kumar	1	0	0.1	1	0
1	1	2008-04-18	M Chinnaswamy Stadium	Kolkata Knight Riders	Royal Challengers Bangalore	BB McCullum	P Kumar	1	0	0.2	1	0
2	1	2008-04-18	M Chinnaswamy Stadium	Kolkata Knight Riders	Royal Challengers Bangalore	BB McCullum	P Kumar	2	0	0.2	2	0
3	1	2008-04-18	M Chinnaswamy Stadium	Kolkata Knight Riders	Royal Challengers Bangalore	BB McCullum	P Kumar	2	0	0.3	2	0
4	1	2008-04-18	M Chinnaswamy Stadium	Kolkata Knight Riders	Royal Challengers Bangalore	BB McCullum	P Kumar	2	0	0.4	2	0

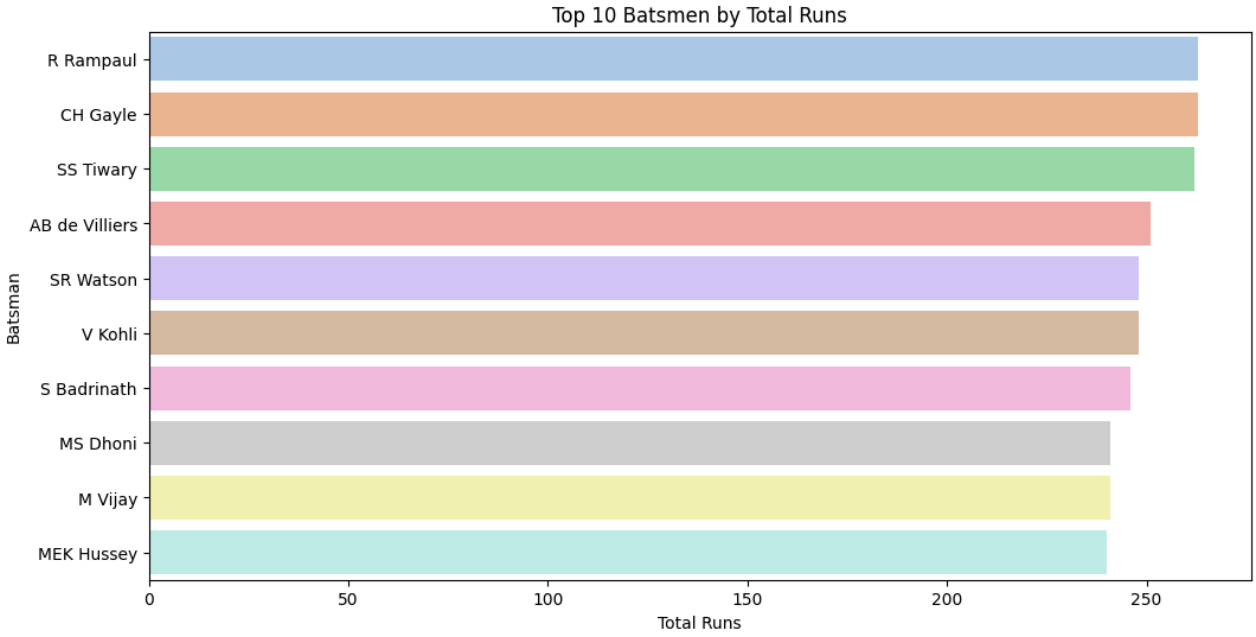
Next steps: Generate code with ipl New interactive sheet

```
data = ipl.copy()
matches_per_venue = data[['mid', 'venue']].drop_duplicates()
matches_count = matches_per_venue['venue'].value_counts()

plt.figure(figsize=(12, 6))
sns.barplot(x=matches_count.values, y=matches_count.index, palette="rainbow")
plt.title('Number of Matches per Venue')
plt.xlabel('Number of Matches')
plt.ylabel('Venue')
plt.show()
```

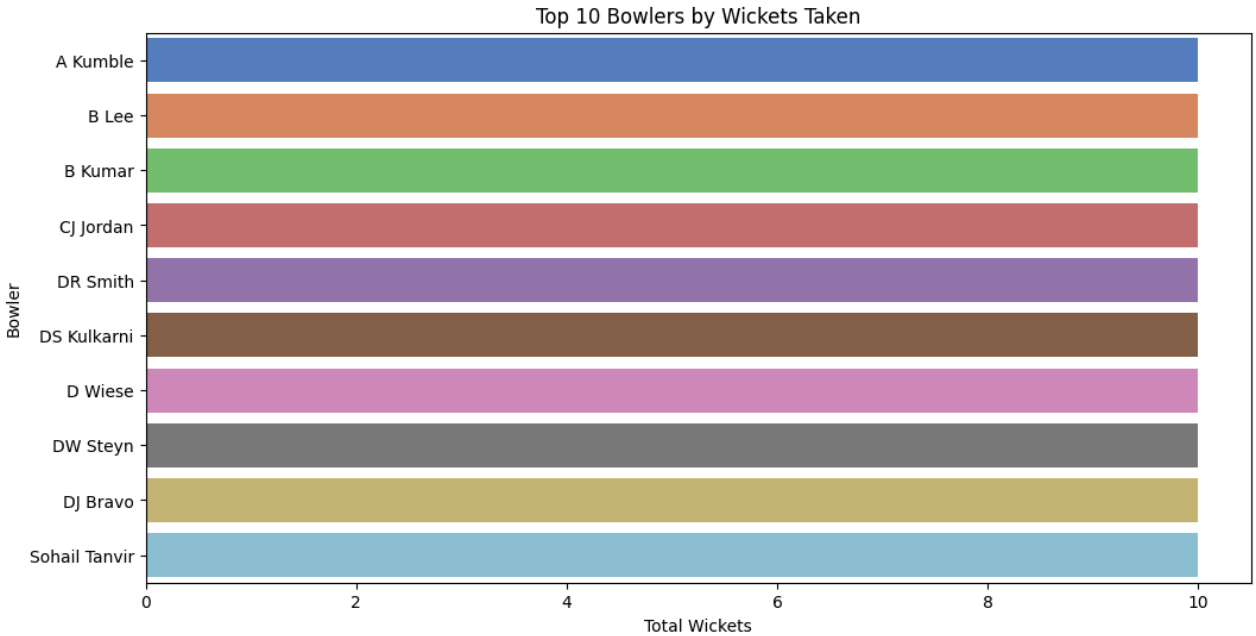



```
runs_by_batsman = data.groupby('batsman')['runs'].max().sort_values(ascending=False).head(10)

plt.figure(figsize=(12,6))
sns.barplot(x=runs_by_batsman.values, y=runs_by_batsman.index,palette="pastel")
plt.title('Top 10 Batsmen by Total Runs')
plt.xlabel('Total Runs')
```

```
plt.ylabel('Batsman')
plt.show()
```



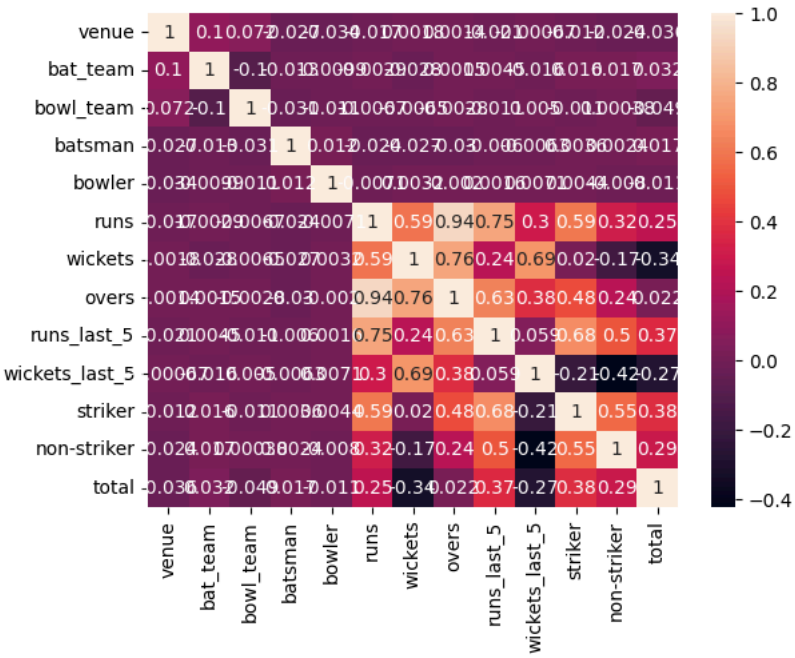

```
wickets_by_bowler = data.groupby(
    'bowler')['wickets'].max().sort_values(ascending=False).head(10)

plt.figure(figsize=(12, 6))
sns.barplot(x=wickets_by_bowler.values,
            y=wickets_by_bowler.index, palette="muted")
plt.title('Top 10 Bowlers by Wickets Taken')
plt.xlabel('Total Wickets')
plt.ylabel('Bowler')
plt.show()
```




```
from sklearn.preprocessing import LabelEncoder

cat_cols = ['bat_team', 'bowl_team', 'venue', "batsman", "bowler"]

data_encoded = data.copy()

label_encoders = {}
```

```
for col in cat_cols:
    le = LabelEncoder()
    data_encoded[col] = le.fit_transform(data_encoded[col])
    label_encoders[col] = le
```

```
data_corr = data_encoded.drop(columns=["date", "mid"], axis=1)
sns.heatmap(data_corr.corr(), annot=True)
plt.show()
```




```
from sklearn.model_selection import train_test_split
feature_cols = ['bat_team', 'bowl_team', 'venue', 'runs',
                'wickets', 'overs', 'striker', 'batsman', 'bowler']

X = data_encoded[feature_cols]
y = data_encoded['total']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
```

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
model = keras.Sequential([
    keras.layers.Input( shape=(X_train_scaled.shape[1],)),
    keras.layers.Dense(512, activation='relu'),
    keras.layers.Dense(216, activation='relu'),
    keras.layers.Dense(1, activation='linear')
])

huber_loss = tf.keras.losses.Huber(delta=1.0)  # You can adjust the 'delta' parameter as needed
model.compile(optimizer='adam', loss=huber_loss)
```

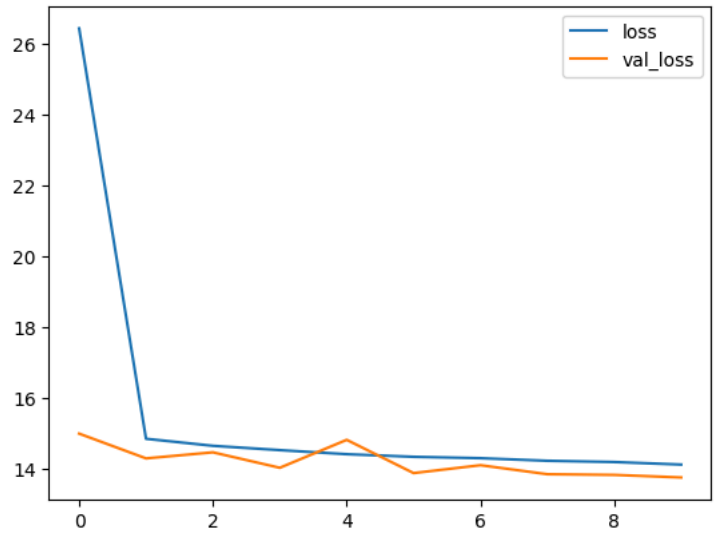
```
model.fit(X_train_scaled, y_train, epochs=10, batch_size=64, validation_data=(X_test_scaled, y_test))
```

```
Epoch 1/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 8s 7ms/step - loss: 50.2874 - val_loss: 15.0081
Epoch 2/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 7s 8ms/step - loss: 14.9861 - val_loss: 14.3098
Epoch 3/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - loss: 14.7712 - val_loss: 14.4809
Epoch 4/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 14.5841 - val_loss: 14.0452
Epoch 5/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 14.3542 - val_loss: 14.8328
Epoch 6/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 14.3192 - val_loss: 13.8959
Epoch 7/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 6s 8ms/step - loss: 14.2966 - val_loss: 14.1180
```

```
Epoch 8/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - loss: 14.2066 - val_loss: 13.8638
Epoch 9/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 7s 8ms/step - loss: 14.3423 - val_loss: 13.8451
Epoch 10/10
832/832 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - loss: 14.2072 - val_loss: 13.7723
<keras.src.callbacks.history.History at 0x7f144b58dd60>
```

```
model_losses = pd.DataFrame(model.history.history)
model_losses.plot()
```

```
<Axes: >
```




```
from sklearn.metrics import mean_absolute_error, mean_squared_error
predictions = model.predict(X_test_scaled)

mean_absolute_error(y_test, predictions)
```

```
713/713 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step
14.262760162353516
```

```
import numpy as np
import ipywidgets as widgets
from IPython.display import display, clear_output
import warnings
warnings.filterwarnings("ignore")

venue = widgets.Dropdown(options=list(
    label_encoders['venue'].classes_), description='Select Venue:')
venue.style = {'description_width': 'initial'}

batting_team = widgets.Dropdown(options=list(
    label_encoders['bat_team'].classes_), description='Select Batting Team:')
batting_team.style = {'description_width': 'initial'}

bowling_team = widgets.Dropdown(options=list(
    label_encoders['bowl_team'].classes_), description='Select Bowling Team:')
bowling_team.style = {'description_width': 'initial'}

striker = widgets.Dropdown(options=list(
    label_encoders['batsman'].classes_), description='Select Striker:')
striker.style = {'description_width': 'initial'}

bowler = widgets.Dropdown(options=list(
    label_encoders['bowler'].classes_), description='Select Bowler:')
bowler.style = {'description_width': 'initial'}

runs = widgets.IntText(value=0, description='Runs:', style={
                       'description_width': 'initial'})
wickets = widgets.IntText(value=0, description='Wickets:', style={
                          'description_width': 'initial'})
overs = widgets.FloatText(value=0.0, description='Overs:', style={
                          'description_width': 'initial'})
striker_ind = widgets.IntText(value=0, description='Striker:', style={
                              'description_width': 'initial'})  # Assuming 0 or 1
predict_button = widgets.Button(description="Predict Score")

output = widgets.Output()


```

```
def predict_score(b):
    with output:
        clear_output()  # Clear previous output

        encoded_venue = label_encoders['venue'].transform([venue.value])[0]
        encoded_batting_team = label_encoders['bat_team'].transform(
            [batting_team.value])[0]
        encoded_bowling_team = label_encoders['bowl_team'].transform(
            [bowling_team.value])[0]
        encoded_striker = label_encoders['batsman'].transform([striker.value])[
            0]
        encoded_bowler = label_encoders['bowler'].transform([bowler.value])[0]

        input_features = [
            encoded_batting_team,
            encoded_bowling_team,
            encoded_venue,
            runs.value,
            wickets.value,
            overs.value,
            striker_ind.value,
            encoded_striker,
            encoded_bowler
        ]

        input_array = np.array(input_features).reshape(1, -1)
        input_array = scaler.transform(input_array)
        predicted_score = model.predict(input_array)

        print(f"Predicted Total Runs: {int(predicted_score[0])}")


predict_button.on_click(predict_score)

display(venue, batting_team, bowling_team, striker, bowler,
        runs, wickets, overs,
        striker_ind,
        predict_button, output)
```

Select Venue: Wankhede Stadium

Select Batting Team: Mumbai Indians

Select Bowling Team: Royal Challengers Bang

Select Striker: RG Sharma

Select Bowler: YS Chahal

Runs: 60

Wickets: 0

Overs: 6

Striker: 1

Predict Score

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 55ms/step
Predicted Total Runs: 180
```